

MoTE: Mixture of Task Experts for Multi-Task Video Understanding

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Abstract

Procedural video-language models must solve heterogeneous tasks from the same visual evidence, including action recognition, forecasting, and procedure prediction. Dense transformer decoders share the same feed-forward networks across tasks, which can entangle task behavior and make controlled capability expansion difficult. Sparse Mixture-of-Experts (MoE) decoders provide conditional computation, but token-level learned routing is not naturally aligned with task-level procedural objectives. We propose **Mixture of Task Experts (MoTE)**, a decoder architecture that converts large language model feed-forward networks into task-specific experts while keeping the multimodal backbone shared. Each example follows one sample-level task route, so active task-expert computation remains independent of the number of stored task experts. We instantiate this design as **VideoLLM-MoTE** and evaluate it on five COIN benchmarks using explicit task routes. The five-expert model activates roughly 2B LLM parameters per sample and achieves higher average top-1 accuracy than recent VideoLLM baselines under the same five-task protocol. Under the same expert topology, it also improves over dense all-expert activation and learned sparse-routing controls. These results show that task-structured routing provides an interpretable and compute-efficient decoder alternative for multi-task video-language learning.

1 Introduction

Intelligent multimodal assistants are expected to follow human instructions over visual observations, from answering what is happening in a procedure to anticipating the next action or producing a concise procedural summary. Instruction tuning has become an effective way to align vision-language models with such diverse user requests, connecting pretrained visual encoders to language models through learned projectors and instruction-following supervision [9, 26, 63]. Video-language models extend this interface to temporal inputs, where a deployed assistant may receive either a complete clip or a live stream of frames [8, 9, 20, 60, 40]. In procedural video understanding, however, the challenge is not only to encode time. The same visual evidence must support heterogeneous tasks such as recognizing the current action, forecasting future steps, identifying the goal, and producing procedure-level responses.

This task diversity creates a concrete modeling tension. Multi-task instruction tuning can improve generalization by exposing a model to many task formats [9, 45], but prior vision-language studies also show that mixing dissimilar tasks can create negative transfer when all

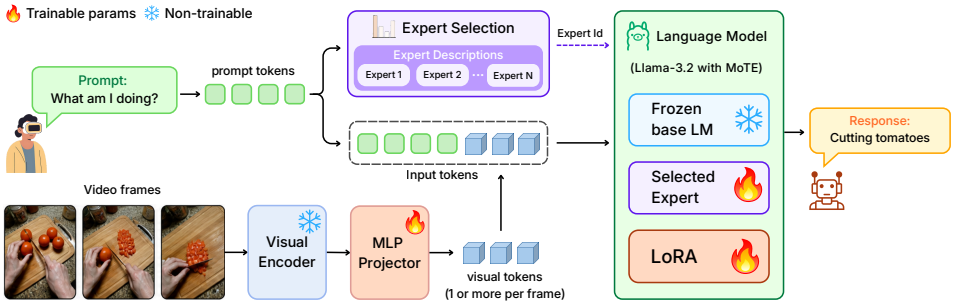


Figure 1: VideoLLM-MoTE overview. A user prompt and sampled video frames are converted into prompt and visual tokens before entering an LLM decoder modified with MoTE. Prompt-conditioned expert selection matches the prompt to registered expert descriptions and passes the selected expert id to the decoder. Given the input tokens and expert id, LLM generates a response. The visual encoder and base language model are non-trainable, while the MLP projector, LoRA adapters, and selected task experts are trainable.

tasks update the same parameters [14]. For procedural videos, task prompts often differ in temporal horizon and output structure even when the underlying frames are shared. A dense decoder therefore has to encode task-specific transformations inside the same feed-forward blocks used by every task.

Modern video-language systems have made progress on long and streaming inputs by improving memory, causal processing, token selection, and computation scheduling [8, 9, 18, 57]. These advances make video inputs more usable, but decoder-side task specialization often remains implicit: the same language-decoder path must express every prompt type. A useful multi-task video-language decoder should therefore preserve shared visual-language grounding while exposing task-specific computation in a controllable part of the model.

Sparse Mixture-of-Experts (MoE) transformers offer a natural starting point for conditional computation [12, 43, 48]. In language, vision, and large vision-language model settings, sparse experts can increase stored capacity while activating only a subset of parameters per input [25, 29]. Standard sparse MoE layers, however, usually route tokens or patches with learned gates. That granularity is effective for scaling, but it does not expose a stable task-level computation path for procedural prompts. Our setting uses known procedural task structure and asks whether the decoder itself can expose task-level routes.

We propose **Mixture of Task Experts (MoTE)**, a decoder architecture that makes task identity an explicit routing variable. MoTE converts each decoder feed-forward network (FFN) into task-specific experts and an always-active shared expert, while keeping the video-language backbone and attention path shared. Each video-prompt example follows one sample-level task route, so active task computation does not grow with the number of stored task experts. Because experts are explicit modules, the architecture can adapt to new tasks by adding task experts while retaining previously learned knowledge in the shared backbone and existing experts. This serves as an architectural principle that we demonstrate is broadly applicable across different domains. During training, task labels provide direct route supervision; at inference time, the route can be supplied externally or selected by matching the prompt to registered expert descriptions.

We evaluate this architecture as VideoLLM-MoTE on five COIN tasks. VideoLLM-

MoTE outperforms recent COIN baselines while using a smaller active-parameter budget. To verify the cross-domain applicability of this architecture, we further conduct tests on document understanding, successfully demonstrating task separation for Key Information Extraction (KIE) and Optical Character Recognition (OCR) across two different datasets. Expert-addition analysis and the explicit expert-removal mechanism further illustrate how task-aligned FFN experts can serve as modular units.

Our contributions are:

1. We introduce MoTE, a modular decoder conversion that replaces shared LLM feed-forward networks with task-routed FFN experts, while retaining shared components for task-agnostic transformations.
2. We instantiate MoTE as VideoLLM-MoTE for multi-task procedural video understanding and show that task-level routing improves over recent VideoLLM baselines, and learned sparse-routing controls under the same expert topology.
3. We show that the same decoder expertization principle transfers beyond video-language understanding by converting GLM-OCR [18] into GLM-OCR-MoTE-0.9B+2E, separating OCR and receipt key information extraction (KIE) task routes.
4. We analyze the modularity of MoTE through expert addition and removal, showing that task experts can be introduced or removed as architectural modules while preserving previously learned knowledge.

2 Related Work

Streaming Procedural Video Understanding. Recent video-language work moves from offline video understanding toward causal, deployable assistants that must process frames as they arrive. VideoLLM-online [8] formalizes streaming dialogue, where a model must decide when to answer during an incoming video stream. VideoLLM-MoD [57] reduces redundant layer computation with mixture-of-depths routing while retaining richer visual tokens. Other systems improve long-context memory, causal temporal modeling, event-triggered reasoning, and always-on interaction [8, 17, 18, 55, 56, 60]. Recent benchmarks further show that online video understanding requires multiple procedural capabilities, including timestamped perception, anticipation, temporal reasoning, and multi-turn assessment [8, 21, 30, 58, 59, 60]. These requirements make procedural video-language modeling inherently multi-task. Existing streaming methods mainly improve memory, token selection, visual encoding, or computation depth. In contrast, MoTE studies how task-specific decoder computation can support heterogeneous procedural objectives while keeping the video-language backbone shared.

Instruction Tuning and Task Conflict. Instruction-tuned vision-language models connect pretrained visual encoders to large language models through lightweight projectors, enabling general-purpose systems such as BLIP-2, LLaVA, and InstructBLIP [8, 26, 53]. Instruction tuning teaches these models to map diverse task templates to appropriate textual responses [45], and visual instruction tuning can improve held-out task generalization [9]. However, this shared-decoder design also creates a multi-task learning problem: heterogeneous prompts, objectives, and output formats compete for the same model capacity. This

Method	Domain	Routing Level	Routing Type	Interpretability	Task-Aware
Sparse MoE [10]	Language	Token	Learned top- k	Low	No
GShard [11]	Language	Token	Learned top- k	Low	No
Switch Transformer [12]	Language	Token	Learned top-1	Low	No
V-MoE [13]	Vision	Patch	Learned top- k	Low	No
Sparse Upcycling [14]	Language/vision	Token	Expert Choice/Learned top- k	Low	No
LLaMA-MoE [15]	Language	Token	Learned top- k	Low	No
MoE-LLaVA [16]	Vision-language	Token	Learned top- k	Low	No
MoCLE [17]	LVLm instruction	Sample	Cluster-conditioned learned top-1	Medium	Yes
M ³ ViT [18]	Multitask vision	Token/Patch	Task-conditioned learned top- k	Medium	Yes
Edge-MoE [19]	Multitask vision	Token/Patch	Task-conditioned learned top- k	Medium	Yes
Cross-task MoE [20]	Language	Sample	Task-representation-based learned top- k	Medium	Yes
Ours (MoTE)	Video-language	Sample	Task index / prompt-conditioned expert selection	High	Yes

Table 1: Comparison of MoE approaches. Prior sparse transformers primarily route tokens or patches with learned gates. Task-level methods make routing more aligned with task semantics by using task representations, or task-conditioned features in the gating function, but they are not designed for modular video-language decoders. MoTE uses sample-level task routing over decoder FFN experts; the reported experiments use explicit task routes, while prompt-description matching provides an expert-selection component for inference.

conflict is well studied in multi-task learning, where shared representations can improve transfer but may also cause negative transfer when task objectives interfere [0, 44, 47, 54, 53]. In procedural video-language modeling, this conflict is sharper because the same visual evidence supports recognition, forecasting, and procedure-level reasoning, yet each task requires different temporal decisions and output formats. MoTE targets this setting by keeping the video-language backbone shared while assigning task-specific transformations to decoder FFN experts.

Sparse MoE and Task-Level Routing. Sparse Mixture-of-Experts transformers add conditional capacity through routed experts. Foundational MoE systems route tokens or image patches with learned gates for language and vision scaling [10, 11, 13, 18], while dense-to-sparse construction reduces the cost of building MoE models by copying or partitioning pre-trained FFNs into experts [14, 15, 16]. These designs scale capacity effectively, but token- or patch-level gates are not naturally aligned with user-facing task boundaries. Task-aware MoE methods move routing closer to task semantics by using task indices, task representations, or task-conditioned features in the routing function. Cross-task MoE uses task representations to select layer-wise expert mixtures across NLP tasks [52], M³ViT uses task-dependent routing for multi-task vision [17], and Edge-MoE focuses on efficient execution for task-aware ViT MoE models [19]. MoCLE addresses task conflict in instruction tuning with LoRA experts conditioned on instruction clusters [12, 19]. MoTE differs by placing specialization inside the LLM decoder FFN path of a video-language model: each video-prompt sample selects one task route, activating the corresponding FFN expert at each decoder layer while attention, the visual encoder, and the MLP projector remain shared. Table 1 summarizes these distinctions across routing level, routing type, interpretability, and task awareness.

3 Method

3.1 Overview

We study multi-task video-language learning when the training set provides a discrete task structure. Each example is a tuple (v, x, t, y) , where v is a video or sampled frame sequence,

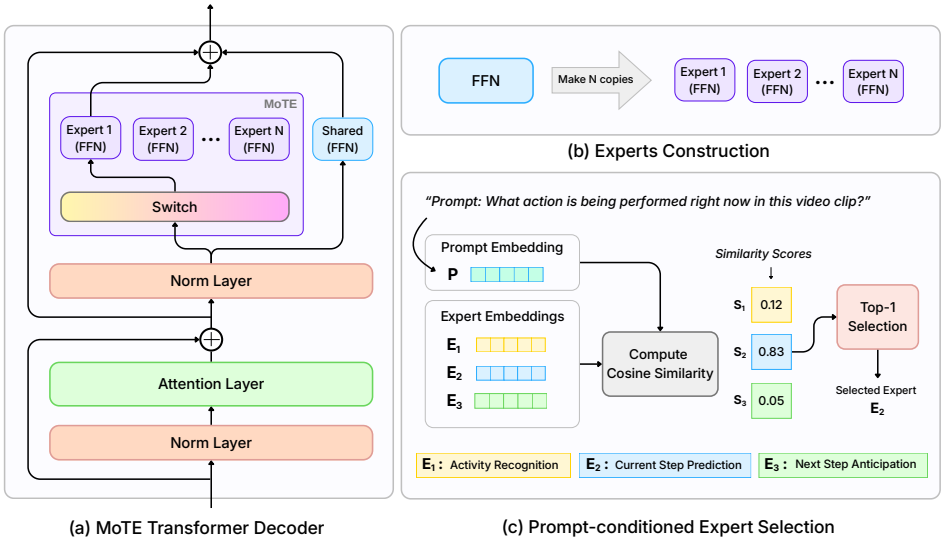


Figure 2: **MoTE decoder and expert selection.** (a) In each decoder layer, attention remains shared, while the FFN layer is replaced by a shared FFN expert and a switch-selected task FFN experts. Both expert outputs are added to the residual stream. (b) Task experts are initialized by copying the pretrained dense FFN into multiple task branches. (c) Prompt-conditioned expert selection embeds the prompt and registered expert descriptions, computes cosine similarities, and selects the top-1 expert for the full video-prompt sample.

x is a natural-language prompt, $t \in \{1, \dots, M\}$ is the task identity, and y is the target response. The learning objective is to model $p_\theta(y | v, x, t)$ while retaining the ability to answer at inference time when the task is either explicitly provided or only implied by the prompt.

As shown in Figure 1, VideoLLM-MoTE consists of a vision encoder, an MLP projector, and a large language model with MoTE FFN expert blocks. We introduce the architecture of VideoLLM-MoTE in Section 3.2, the training objective in Section 3.3, and modular task expert adaptation in Section 3.4.

3.2 Architecture of VideoLLM-MoTE

Shared Backbone. The shared backbone follows the standard video-language pattern shown in Figure 1. A pretrained vision encoder maps the sampled frames in v to a visual token sequence $Z = [z_1, \dots, z_P] \in \mathbb{R}^{P \times C}$, where P is the total number of visual tokens and C is the visual feature dimension. A trainable two-layer MLP projector f_{proj} maps these features into the LLM hidden size D , giving $U_v = f_{\text{proj}}(Z) \in \mathbb{R}^{P \times D}$. The prompt x is embedded by the language model token embedding layer, yielding $U_x \in \mathbb{R}^{N \times D}$ for N text tokens. The chat template marks visual positions with $\langle v \rangle$; after projection, those placeholders are replaced by U_v , producing the decoder input sequence $H_0 = \text{Template}(U_x, U_v)$, which concatenates visual and text token embeddings. All samples share this visual encoder, MLP projector, language token embedding layer, and attention path. Further implementation details such as frame rate, resolution, visual-token count, and backbone scale are reported in Section 4.1.

MoTE decoder blocks. MoTE modifies the transformer decoder layer in the LLM. Only the FFN sublayer is changed, so task-dependent transformations can be selected without duplicating the full video-language model. Concretely, let the dense decoder contain L layers. In layer l , the original FFN is denoted by F_l . As shown in Figure 2(a), MoTE replaces F_l with a set of task experts

$$\mathcal{F}_l = \{F_l^1, F_l^2, \dots, F_l^M\}$$

and a shared expert F_l^{sh} . Each task expert has the same architecture as the original FFN and is associated with one task. The shared expert is active for every task and captures transformations that should remain common, while the selected task expert specializes to task-specific supervision and output formats.

For hidden states H_l , the shared pre-norm attention block produces

$$\tilde{H}_l = H_l + \text{GQA}_l(\text{LN}(H_l)),$$

where GQA_l denotes grouped-query attention and $\text{LN}(\cdot)$ denotes layer normalization. Given a task route $r \in \{1, \dots, M\}$, the expert block computes

$$H_{l+1} = \tilde{H}_l + \lambda_{\text{sh}} F_l^{\text{sh}}(\text{LN}(\tilde{H}_l)) + \lambda_{\text{task}} F_l^r(\text{LN}(\tilde{H}_l)).$$

The coefficients λ_{sh} and λ_{task} are expert scaling factors; in our runs the routed scaling factor is 1.0. As illustrated in Figure 2(b), both routed and shared experts are initialized as copies of the pretrained dense FFN, following the dense-to-expert initialization principle used in sparse upcycling [23]. This avoids starting task experts from random transformations and allows the shared and task-specific paths to diverge during training.

This construction separates two roles that are entangled in a dense decoder. The shared expert learns common video-language transformations, while F_l^m can specialize to the supervision and output format of task m . Because only one task expert is activated for a sample, the number of stored experts can grow with the task set without requiring all experts to run for every input.

Task-route selection. MoTE routes the full video-prompt sample rather than individual tokens: all visual and textual tokens in an example pass through the same task expert at every decoder layer. This aligns computation with the user-facing task and avoids a learned-gate or load-balancing loss in the explicit task-index setting. For a training tuple (v, x, t, y) , where v is the video input, x is the prompt, and $t \in \{1, \dots, M\}$ is the task identity among M task experts, the direct route is $r_{\text{idx}}(v, x, t) = t$.

During joint training, this known task identity provides route supervision. Only the selected task expert and the shared expert receive gradients for a given example, exposing each task expert to a consistent task distribution.

At inference time, the task identity may be unavailable. As illustrated in Figure 2(c), each expert index $m \in \{1, \dots, M\}$ is associated with a short natural-language description d_m , such as “predict the next step” or “identify the overall procedure”. Let $\phi(\cdot)$ be a frozen text encoder, and let $\bar{\phi}(z) = \phi(z) / \|\phi(z)\|_2$ denote the normalized embedding of any text string z . We infer a prompt-conditioned route by nearest-neighbor matching between the prompt and the expert descriptions:

$$r_{\text{prompt}}(x) = \arg \max_{m \in \{1, \dots, M\}} \bar{\phi}(x)^\top \bar{\phi}(d_m).$$

This selector is useful for modular deployment. In contrast to a learned gate whose output dimension is fixed by the experts observed during training, a description registry can add or remove natural-language expert descriptions while leaving the decoder parameters and remaining expert routes unchanged. We therefore use prompt-conditioned selection as an architectural component for inference when a task index is unavailable.

3.3 Training Objective

Following VideoLLM-online [4], the training objective supervises two behaviors in the same autoregressive video-language sequence. At response tokens, the model predicts the next answer token under the selected expert route. At streaming positions that don't require a response, it predicts EOS so it stays silent until needed. Let $p_\theta(\cdot | H_{\leq j}, r)$ denote the next-token distribution after prefix $H_{\leq j}$ under route r . We use two binary indicators: $l_j = 1$ when position j is a supervised language-response token, and $s_j = 1$ when position j is a frame position assigned streaming silence supervision. The loss is

$$\mathcal{L} = \frac{1}{N} \sum_{j=1}^N \underbrace{(-l_{j+1} \log p_\theta(a_{j+1} | H_{\leq j}, r))}_{\text{LM Loss}} - \underbrace{ws_j \log p_\theta(\text{EOS} | H_{\leq j}, r)}_{\text{Stream Loss}},$$

where a_{j+1} is the next target token, w balances the streaming term, and N is the sequence length. We use $w = 1$ in reported runs. In our implementation, s_j is nonzero only for the last visual token of a frame when that frame is not immediately followed by a supervised language response. For all COIN offline benchmark tasks, stream messages are marked as non-learned positions, so $s_j = 0$ for frame tokens and the loss reduces to standard routed language modeling:

$$\mathcal{L}_{\text{COIN}} = -\frac{1}{N} \sum_{j=1}^N l_{j+1} \log p_\theta(a_{j+1} | H_{\leq j}, r).$$

3.4 Modular Task Expert Adaptation

Beyond joint training, MoTE supports task adaptation by changing the task-specific FFN experts associated with the affected capability. New tasks can be introduced by attaching new routed experts, while tasks that are no longer needed can be removed from the active expert set without rewriting the shared video-language backbone or the remaining task experts.

Expert addition. Assume a model has been trained for tasks $\{1, \dots, K\}$. To add task $K+1$, we insert a new expert F_l^{K+1} into every decoder layer and initialize it from the corresponding dense FFN or the shared expert. We then train on the new task using route $r = K+1$ while freezing the visual encoder, MLP projector, attention blocks, shared experts, and existing task experts. Only the new experts are updated:

$$\Theta_{\text{train}} = \{\theta(F_l^{K+1}) : l = 1, \dots, L\}.$$

This isolates new-task learning from previously learned task modules: old routes keep the parameters that produced their earlier behavior, which helps mitigate catastrophic forgetting during expansion.

Method	Not use HowTo100M	COIN Benchmark Top-1 Accuracy↑					Avg.
		Step	Next	Proc.	Task	Proc.+	
<i>Non-LLM procedural video methods</i>							
ClipBERT [24]	✓	30.8	–	–	65.4	–	–
TimeSformer [9]	✗	46.5	34.0	17.0	85.3	40.1	44.6
Paprika [62]	✗	51.0	43.2	–	85.8	–	–
DistantSup [60]	✗	54.1	39.4	–	90.0	41.3	–
VideoTF [64]	✗	56.5	42.4	40.2	91.0	46.4	55.3
ProcedureVRL [66]	✗	56.9	46.8	–	90.8	–	–
VideoTaskGraph [9]	✗	57.2	40.2	–	90.5	–	–
<i>LLM-based video-language methods</i>							
VideoLLM-online-7B-v1 [8]	✓	59.8	48.1	47.9	92.1	52.9	60.2
VideoLLM-online-8B-v1+ [8]	✓	63.1	49.1	49.8	92.7	54.1	61.8
VideoLLM-online-1B-v1+	✓	57.7	47.4	48.1	92.1	50.8	59.3
VideoLLM-MoD [67]	✓	63.4	49.7	49.8	92.8	53.3	61.8
VideoLLM-MoTE-1B+5E (ours)	✓	65.1	50.3	50.5	94.4	54.2	62.9

Table 2: Comparison with prior COIN baselines. *Not use HowTo100M* indicates whether the method avoids HowTo100M pretraining. The average is reported when all five task scores are available. Our method achieves 62.9% average accuracy outperforming other baselines.

Expert removal. When a task is outside the target deployment domain, or when a capability should be withdrawn for unlearning or compliance reasons, its experts can be dropped from every decoder layer and its description can be removed from the selector. This gives MoTE a direct capability-removal mechanism that is difficult to express in a dense decoder, where task behavior is distributed across shared parameters.

4 Experiments

4.1 Experimental Setup

Datasets and tasks. Following VideoLLM-online [8] and VideoLLM-MoD [67], we evaluate VideoLLM-MoTE in both offline procedural benchmarks and an online narration setting. For offline evaluation, we use COIN [60], an instructional-video dataset with temporally annotated procedural steps. We report five COIN benchmark tasks: *Step* recognition, *Next* step forecasting, *Task* recognition, *Proc.* procedure forecasting, and *Proc.+* task-conditioned procedure forecasting. We use VideoLLM-MoTE-1B+5E for this setting, with one routed expert for each COIN task. To evaluate task extension across datasets, we further use the Ego4D narration stream [16], where timestamped egocentric narrations define when the model should speak and what it should say. We first train an Ego4D narration route and then attach five COIN routes trained on the COIN tasks. The resulting VideoLLM-MoTE-1B+6E is evaluated as a single expanded model that preserves the original online narration capability while adding offline procedural reasoning tasks. Finally, we test whether the same MoTE method transfers beyond video understanding. We convert GLM-OCR-0.9B into GLM-OCR-MoTE-0.9B+2E by retaining an OCR expert and adding a receipt key information extraction (KIE) expert. We evaluate direct image-to-structured-field extraction on SROIE [70] and CORD [40], while also measuring whether the original OCR route is preserved after adding the KIE route.

Model	LLM	LLM cfg.			Experts		Params (B)		COIN Avg.↑
		Layers	Width	FFN	T	S	Total	Active	
VideoLLM-online-7B-v1 [8]	Llama-2-7B	32	4096	11008	–	–	6.7	6.7	60.2
VideoLLM-online-8B-v1+ [8]	Llama-3-8B	32	4096	14336	–	–	8.0	8.0	61.8
VideoLLM-MoD [14]	Llama-3-8B	32	4096	14336	–	–	8.0	8.0	61.8
VideoLLM-online-1B-v1+	Llama-3.2-1B	16	2048	8192	–	–	1.2	1.2	59.3
VideoLLM-MoTE-1B+5E (ours)	Llama-3.2-1B	16	2048	8192	5	1	5.3	2.0	62.9

Table 3: LLM-only parameter summary and COIN average accuracy for VideoLLM baselines and VideoLLM-MoTE. Parameter counts include token embeddings and transformer decoder layers, but exclude visual encoders and MLP projectors. Dense baselines do not decompose the decoder into routed experts. VideoLLM-MoTE-1B+5E achieves the highest COIN average while activating only 2.0B LLM parameters per sample.

Implementation details. All reported VideoLLM-MoTE models are trained on four A100-80GB GPUs with videos sampled at 2 FPS and capped at 1200 frames. Our default COIN setting uses a SigLIP 2 ViT-L/16-384 visual encoder [13], a two-layer MLP projector, and a Llama-3.2-1B-Instruct decoder [14]. Given our limited computational resources, we select this 1B decoder as the default backbone rather than training larger variants; this keeps the study focused on routing behavior under a tractable training budget. Each frame is represented by one CLS token plus a 3×3 pooled spatial grid, giving 10 visual tokens per frame. We fine-tune the language model with LoRA [19] and keep the MLP projector trainable. The COIN model contains five routed task experts and one shared expert; each sample activates only the selected routed expert and the shared expert. Supplementary Sections A.1 and A.2 provide detailed architecture specifications and training hyperparameters.

Evaluation Metrics. For COIN, we follow the generated-answer evaluation used by prior procedural video-language work [6, 12]: model outputs are decoded as text, normalized, mapped back to the task label space, and scored as top-1 accuracy. For *Proc.* and *Proc.+*, the generated numbered list is evaluated step-wise over the aligned future-action positions. We report the arithmetic mean over the five COIN tasks when all task scores are available. For Ego4D narration validation, following VideoLLM-online, we report perplexity (PPL) for language modeling quality, TimeDiff for temporal response alignment, and Fluency as a combined language-and-timing score. Lower is better for PPL and TimeDiff, while higher is better for Fluency. We refer readers to [8] for the detailed definition and computation of the online narration metrics.

4.2 Comparison with Prior COIN Baselines

Table 2 compares VideoLLM-MoTE-1B+5E against prior COIN baselines under the same five-task protocol. The upper block contains earlier non-LLM procedural-video methods, while the lower block contains LLM-based VideoLLM variants. VideoLLM-MoTE-1B+5E achieves the highest accuracy on all five COIN tasks, improving the five-task average from 61.8% for the strongest 8B VideoLLM baseline to 62.9%.

The public 7B–8B baselines differ from our model in decoder scale, visual encoder, and training recipe. We therefore treat them as protocol-level comparisons rather than isolated architecture controls. Under this conservative interpretation, VideoLLM-MoTE-1B+5E remains competitive: despite using a smaller active LLM path, it exceeds the reported average

accuracy of the larger VideoLLM baselines.

To reduce the scale confound, we also include VideoLLM-online-1B-v1+, a VideoLLM-online baseline trained from the same Llama-3.2-1B and SigLIP-family initialization as our default model. In this comparison, the MoTE conversion increases the five-task average from 59.3% to 62.9%. The controlled evidence for the MoTE decoder design therefore comes from two comparisons: the matched 1B VideoLLM-online baseline in Table 2, and the fixed-topology routing controls in Table 8.

Table 3 makes the active-computation advantage explicit. The 8B VideoLLM baselines activate the full 8.0B-parameter LLM for each sample, whereas VideoLLM-MoTE-1B+5E activates only a 2.0B LLM path through the selected task expert and shared expert. Thus, the reported COIN average is higher despite using roughly one quarter of the active LLM parameters of the strongest 8B baselines.

4.3 Modular Task Adaptation via Expert Addition and Removal

Table 4 evaluates modular task adaptation through sequential expert addition and removal. In the addition phase, we first train the *Step* route and then introduce the *Next*, *Proc.*, *Task*, and *Proc.* + routes one at a time. At each expansion stage, only the newly added task route is trained, while previously trained routes are kept fixed. Consequently, earlier task accuracies remain unchanged across later stages, showing that adding a new task does not overwrite existing task-specific modules. The final addition stage reaches 61.0% average accuracy over the five active tasks, compared with 62.9% for the jointly trained VideoLLM-MoTE-1B+5E. This gap reflects a stability-plasticity trade-off: frozen expert addition prioritizes preservation of existing task behavior, whereas joint training allows shared components and task experts to adapt together for higher peak multi-task accuracy. In the removal phase, dropping an expert disables the corresponding task route, while retained routes keep their parameters and measured accuracies. This supports expert removal as a practical deployment mechanism for disabling task-specific capabilities, although it does not constitute a full unlearning evaluation, which would require testing whether removed-task behavior can still be recovered through other routes or prompts.

4.4 Cross-Dataset Task Extension

Table 5 evaluates cross-dataset task extension in a single model. We start from an Ego4D narration model, which contains an online route for deciding when to speak and what narration to generate. We then expand the same model with five COIN experts and train only the newly added COIN routes. The shared video-language backbone, including the visual connector and decoder attention layers, is not retrained on COIN. Thus, the COIN experts operate on shared representations learned from Ego4D narration training, while only the task-specific decoder FFN routes are adapted to the new COIN objectives. This protocol tests whether a model trained for one dataset-specific task family can acquire a new set of procedural tasks while retaining the original online narration capability in the same checkpoint.

The split baseline rows clarify the evaluation scope. VideoLLM-online and VideoLLM-MoD report strong COIN and Ego4D results, but these scores are obtained from separate task-specific finetuned models rather than from one model that supports both task families. By contrast, VideoLLM-MoTE-1B+6E reports COIN and Ego4D metrics from a single expanded checkpoint that retains the Ego4D narration expert while adding five COIN experts. The expanded model remains competitive on the newly added COIN tasks and preserves

COIN Benchmark	Stage 1 Step	Expert addition stages				Expert removal stages			
		➕ Next	➕ Proc.	➕ Task	➕ Proc.+	✖ Proc.+	✖ Task	✖ Proc.	✖ Next
Step	63.1	63.1	63.1	63.1	63.1	63.1	63.1	63.1	63.1
Next	—	46.2	46.2	46.2	46.2	46.2	46.2	46.2	—
Proc.	—	—	48.4	48.4	48.4	48.4	48.4	—	—
Task	—	—	—	92.8	92.8	92.8	—	—	—
Proc.+	—	—	—	—	54.6	—	—	—	—

Table 4: **Modular task adaptation through sequential expert addition and removal.** We first train a single *Step* expert, then add one task expert at a time for *Next*, *Proc.*, *Task*, and *Proc.+*. After all five experts are available, we remove them in reverse order. Accuracy is reported after each addition or removal stage, and a dash indicates that the corresponding task expert is not present. Stable scores for previously added tasks indicate that adding or removing other experts does not alter their routed inference paths.

Method	Model instance	COIN Benchmark Top-1 Accuracy↑					Ego4D Narration Stream Validation		
		Step	Next	Proc.	Task	Proc.+	PPL↓	TimeDiff↓	Fluency↑
<i>Separate finetuned models reported by prior work</i>									
VideoLLM-online-7B-v1 [9]	COIN-only	59.8	48.1	47.9	92.1	52.9	—	—	—
VideoLLM-online-7B-v1 [9]	Ego4D-only	—	—	—	—	—	2.43	2.25	42.6
VideoLLM-online-8B-v1+ [9]	COIN-only	63.1	49.1	49.8	92.7	54.1	—	—	—
VideoLLM-online-8B-v1+ [9]	Ego4D-only	—	—	—	—	—	2.40	2.05	45.3
VideoLLM-MoD [10]	COIN-only	63.4	49.7	49.8	92.8	53.3	—	—	—
VideoLLM-MoD [10]	Ego4D-only	—	—	—	—	—	2.41	2.04	45.2

VideoLLM-MoTE-1B+6E expert expansion in a single model

VideoLLM-MoTE-1B+6E (ours)	Single expanded model	64.9	47.6	50.5	93.9	56.4	2.57	2.25	51.0
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Table 5: **Cross-dataset task extension from online narration to offline procedural tasks.** Prior VideoLLM-online and VideoLLM-MoD results are reported from separate COIN-only and Ego4D-only finetuned model instances. In contrast, the final row evaluates one expanded VideoLLM-MoTE-1B+6E checkpoint that starts from an Ego4D narration route and adds five COIN task routes. Ego4D validation reports PPL, TimeDiff, and Fluency.

online narration behavior, achieving the highest Fluency value in the table. The Ego4D-only baselines still obtain lower PPL and TimeDiff, indicating better standalone narration likelihood and timing. This result suggests that expert expansion can reuse a frozen shared backbone when the initial training domain and the added tasks have sufficient semantic overlap. Therefore, the comparison should be interpreted as evidence for backward-compatible task expansion across related video-language datasets, not as a direct claim of superior Ego4D-only narration quality or unrestricted transfer to unrelated domains.

4.5 Auxiliary Cross-Domain Evaluation

While our primary experiments focus on procedural video, we include an auxiliary evaluation on Key Information Extraction (KIE) from receipt images to verify if the MoTE FFN conversion mechanism applies outside video tokens. Unlike video benchmarks, this task requires parsing text-rich, static documents into structured outputs. We adapted the dense GLM-OCR-0.9B [11] backbone into the MoTE framework by introducing a shared expert alongside two task-specific routes: a retained Optical Character Recognition (OCR) route for the original capabilities and a newly trained KIE route. The KIE expert was trained on the SROIE and CORD datasets to generate structured field predictions directly from raw document images without an intermediate text-extraction pipeline.

The converted GLM-OCR-MoTE-0.9B+2E model achieves 87.90% Micro-F1 on SROIE

Method	Input Modality	Micro-F1 (%) $\uparrow$
LayoutLLM-7B [10]	Image+Text+Layout	72.12
DocKylin [10]	Image + Text	69.60
Qwen-VL-7B [9]	Image + Text	58.59
LLaVA-1.5-7B [10]	Image + Text	3.83
LLaVAR-7B [10]	Image + Text	2.38
GLM-OCR-0.9B [10]	Image	55.04
GLM-OCR-MoTE-0.9B+2E (ours)	Image	87.90

Table 6: SROIE KIE Micro-F1 comparison for image-using models. GLM-OCR-MoTE-0.9B+2E reaches 87.90%, compared with 72.12% for the strongest image-using baseline in this compact table.

KIE and 95.79% Micro-F1 on CORD, successfully raising the dense GLM-OCR CORD and SROIE baseline from 20.07% and 55.04% respectively to demonstrate effective adaptation to structured extraction. Crucially, the retained OCR route perfectly preserves the original model’s performance on OmniDocBench v1.5 after conversion and KIE training. Full performance metrics are provided in Supplementary Tables 13 and 14.

By leveraging the MoTE architecture, the compact 0.9B GLM backbone is transformed into a highly capable KIE model while retaining perfect OCR scores along its original route. The resulting GLM-OCR-MoTE-0.9B+2E proves highly competitive with several open multimodal models, demonstrating that task-specific experts can successfully attach to non-video backbones to introduce entirely new capabilities without inducing catastrophic forgetting. We present these findings primarily as auxiliary evidence of the architecture’s cross-domain versatility; establishing a broader domain-generalization claim.

4.6 Ablation Study

4.6.1 Effect of Routing Granularity

Table 8 compares how the same five-expert topology behaves under different routing choices. The dense control removes sparsity by activating every routed expert for every sample. The token-level control learns an independent top-1 expert for each token. The sample-level control learns one top-1 expert for the full video–prompt sample, matching MoTE at the sample granularity but without explicit task supervision. MoTE instead uses the known task identity to select the task-aligned expert. The experiment therefore tests whether task-aligned sample routing is more effective than dense expert fusion or learned sparse routing at token and sample granularity.

With the same converted decoder capacity, VideoLLM-MoTE-1B+5E achieves 62.9% average accuracy, compared with 61.5% accuracy of VideoLLM-1B+5E-Dense for dense all-expert activation and 61.4% for token routing. This indicates that task structure is useful for choosing the FFN transformation: activating all routed experts dilutes specialization, while token-level routing breaks the task boundary across the generated sequence. The advantage is especially clear on procedure-oriented tasks, where the requested output depends on a longer temporal horizon and a more structured response format.

The sample-level baseline is closest to MoTE in routing granularity because both choose one expert route for the whole video–prompt sample. The difference is route semantics: sample-level MoE learns a latent top-1 route, whereas MoTE routes by task identity. Sample-

Method	Input Modality	Micro-F1 (%) $\uparrow$
UDOP [10]	Image+Text+Layout	97.60
LayoutLMv2 [10]	Image+Text+Layout	96.01
DocTr [10]	Image+Text+Layout	94.40
Donut [10]	Image + Text	91.60
GLM-OCR-0.9B [10]	Image	20.07
GLM-OCR-MoTE-0.9B+2E (ours)	Image	95.79

Table 7: CORD KIE Micro-F1 comparison for image-using models. GLM-OCR-MoTE-0.9B+2E raises the GLM-OCR baseline from 20.07% to 95.79%, approaching the strongest multimodal baseline while retaining the OCR route discussed in the text.

Method	COIN Benchmark Top-1 Accuracy↑					Avg.
	Step	Next	Proc.	Task	Proc.+	
VideoLLM-1B+5E-Dense	64.4 (0.0)	49.8 (0.0)	48.7 (0.0)	93.2 (0.0)	51.5 (0.0)	61.5 (0.0)
VideoLLM-MoE-Top1-token	63.7 (-0.7)	49.5 (-0.3)	48.5 (-0.2)	93.8 (+0.6)	51.7 (+0.2)	61.4 (-0.1)
VideoLLM-MoE-Top1-sample	64.9 (+0.5)	50.3 (+0.5)	49.2 (+0.5)	93.6 (+0.4)	52.3 (+0.8)	62.1 (+0.6)
VideoLLM-MoTE-1B+5E (ours)	65.1 (+0.7)	50.3 (+0.5)	50.5 (+1.8)	94.4 (+1.2)	54.2 (+2.7)	62.9 (+1.4)

Table 8: **Effect of Routing Granularity.** Numbers in parentheses are accuracy changes relative to the VideoLLM-1B+5E-Dense control; green and red indicate positive and negative changes, respectively. VideoLLM-MoE-Top1-token learns token-level routes, VideoLLM-MoE-Top1-sample learns one route for the full sample, and VideoLLM-MoTE-1B+5E uses explicit task-index routing.

Method	MoE Construction	COIN Benchmark Top-1 Accuracy↑					Avg.
		Step	Next	Proc.	Task	Proc.+	
VideoLLM-MoTE-8B+5E-Split	Splitting FFN	63.3	48.8	48.3	92.9	48.3	60.3
VideoLLM-MoTE-1B+5E (default)	Copying FFN	65.1	50.3	50.5	94.4	54.2	62.9

Table 9: **Effect of Expert Construction Method.** VideoLLM-MoTE-8B+5E-Split partitions each VideoLLM FFN into six equally sized expert groups (one shared expert and five task experts), while VideoLLM-MoTE-1B+5E initializes each shared and task expert as a full copied FFN. Copying FFN layer into multiple experts achieves better performance.

level routing improves over dense activation, but MoTE gives the strongest average accuracy and the largest gains on structured procedure prediction. This suggests that, for task-structured procedural video understanding, MoTE is most effective when the route is aligned with the user-facing task.

4.6.2 Effect of Expert Construction Method

Table 9 reports compares the two expert construction methods. Instead of copying the full pretrained FFN into each expert, **VideoLLM-MoTE-8B+5E-Split** uses the same 8B VideoLLM backbone and partitions each FFN into six equally sized experts, corresponding to one shared expert and five task experts, following LLaMA-MoE [68]. This keeps the experts disjoint inside each FFN rather than giving each task route a full copied FFN.

VideoLLM-MoTE-8B+5E-Split reaches 60.3% average accuracy, below the 62.9% average of the full-FFN-copy VideoLLM-MoTE-1B+5E. This suggests that task-routed video-language experts benefit from retaining the full pretrained FFN transformation in each route before specialization. Splitting neurons into small disjoint task groups is parameter efficient, but it reduces the per-task FFN capacity available for procedural output formats and long-horizon prediction.

4.7 Discussion

The experiments support a simple claim: task identity is a useful and actionable structure for multi-task learning across diverse domains. Under the same expert topology, explicit task-index routing gives the strongest average accuracy across our tested benchmarks among the internal controls and uses the expert capacity more effectively than dense all-expert activation or learned one-expert sparse routing. The clearest gains appear on tasks with highly specialized objectives, where a single shared transformation is more likely to mix incompati-

ble task requirements. This suggests that task routing helps most when task boundaries align with genuinely different computational needs.

The results also clarify where specialization occurs. In MoTE, attention and foundational representation alignment remain shared, while task-specific transformations are assigned to the feed-forward experts. This separates shared context modeling from task-dependent computation: the shared path preserves common multimodal knowledge, while routed experts capture output formats and reasoning patterns specific to each task. The design therefore sits between full parameter sharing and full model duplication.

Finally, the expert-addition and cross-dataset experiments establish the main tradeoff between stability and plasticity. New experts can be added without overwriting previous routes, whether expanding within video benchmarks (from Ego4D to COIN) or adapting to entirely different domains. The auxiliary document-image experiment using the GLM-OCR-0.9B backbone confirms that the same FFN conversion can successfully attach a structured-output route outside of video. By introducing a receipt KIE route alongside the retained OCR route, the model raises CORD micro-F1 from 20.07% to 95.79% from raw images while preserving the reported OmniDocBench OCR metrics in this controlled check. This result reinforces the robustness of the modularity mechanism. Taken together, these multi-domain results indicate that MoTE is most useful when a model must support strong task performance, backward compatibility, and modular expansion in the same architecture.

5 Conclusion

In this work, we introduce MoTE, a modular architecture for multi-task multimodal understanding that decouples task-specific computation from shared foundational knowledge.

The framework converts an LLM decoder’s feed-forward blocks into task-specialized experts surrounding a shared backbone, leveraging task identity for targeted specialization and expansion. We validate this approach across diverse domains. In video understanding, VideoLLM-MoTE surpasses dense all-expert activation, learned sparse-routing controls, and larger VideoLLM baselines on five COIN benchmarks, all while maintaining a smaller active parameter footprint. In document analysis, we demonstrate that MoTE can master new structured-output tasks while perfectly retaining foundational OCR capabilities. Crucially, expert-addition experiments confirm that novel experts integrate seamlessly without overwriting existing routes, enabling continual adaptation and avoiding the interference typical of monolithic decoders.

MoTE assumes that the target task set has meaningful and predefined boundaries. When tasks overlap substantially, or when prompts are ambiguous or compositional, sample-level routing may select an unsuitable expert. Future work should relax this assumption through dynamic task discovery, soft hierarchical routing, or expert composition, enabling open-ended user requests without requiring explicit expert registration.

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A Supplementary Material

The supplementary material is organized as follows. Section A.1 gives parameter accounting for the video-language model. Section A.2 collects the training configuration and the route-description registry. Section A.3 explains the data sampling strategy. Section A.4 gives the rendered prompt templates on a separate page. Section A.5 defines the evaluation protocols, Section A.6 reports an expert-swap diagnostic, Section A.7 visualizes the auxiliary document pipeline, Section A.8 provides the complete auxiliary document-KIE tables, and Section A.9 summarizes scope limitations.

A.1 VideoLLM-MoTE Parameter Details

We separate the video-language model into the SigLIP 2 visual encoder, the two-layer MLP projector, and the Llama decoder:

$$P_{\text{VLM}} = P_{\text{vis}} + P_{\text{proj}} + P_{\text{LLM}}.$$

The main-paper scale table reports the LLM term because the compared VideoLLM base-lines use different visual encoders and connectors; the full model counts below add the visual and projector terms explicitly.

Two conventions are used throughout this parameter summary. *Total* parameters count every stored expert after conversion, whereas *active* parameters count only the modules executed for a single routed sample. The visual encoder and projector are shared by all routes, so they are included once in the full video-language count and do not change with the number of task experts.

SigLIP 2 ViT-L/16-384 Visual Encoder. For the visual encoder, let the image resolution be $R = 384$, patch size $p = 16$, input channels $C_{\text{in}} = 3$, visual width $D_v = 1024$, visual FFN width $F_v = 4096$, and number of visual Transformer layers $L_v = 24$. The number of patch positions is

$$N_p = (R/p)^2 = (384/16)^2 = 576.$$

The patch embedding is a convolution with one D_v -dimensional output vector per $p \times p$ RGB patch, and the encoder uses learned absolute position embeddings:

$$P_{\text{patch}} = C_{\text{in}} p^2 D_v + D_v = 3 \cdot 16^2 \cdot 1024 + 1024 = 787,456,$$

$$P_{\text{pos}} = N_p D_v = 576 \cdot 1024 = 589,824.$$

Each SigLIP 2 vision block contains four attention projections, a two-layer FFN, and two layer norms. With linear biases and LayerNorm scale and bias parameters, one layer has

$$P_{\text{attn}}^v = 4(D_v^2 + D_v),$$

$$P_{\text{ffn}}^v = (D_v F_v + F_v) + (F_v D_v + D_v),$$

$$P_{\text{norm}}^v = 4D_v,$$

$$P_{\text{layer}}^v = P_{\text{attn}}^v + P_{\text{ffn}}^v + P_{\text{norm}}^v = 12,596,224.$$

The encoder stack therefore contributes $L_v P_{\text{layer}}^v = 302,309,376$ parameters. The visual encoder also contains a final LayerNorm and the SigLIP pooling head. The pooling head

1104 consists of one learned probe vector, multi-head attention pooling, a LayerNorm, and a two-
1105 layer MLP:

$$\begin{aligned}
 1106 \quad P_{\text{post}}^v &= 2D_v = 2,048, \\
 1107 \quad P_{\text{pool}}^v &= D_v + 4(D_v^2 + D_v) + 2D_v + (D_v F_v + F_v) + (F_v D_v + D_v) \\
 1108 \quad &= 12,595,200.
 \end{aligned}$$

1110 Thus the visual encoder used by VideoLLM-MoTE has
1111

$$\begin{aligned}
 1112 \quad P_{\text{vis}} &= P_{\text{patch}} + P_{\text{pos}} + L_v P_{\text{layer}}^v + P_{\text{post}}^v + P_{\text{pool}}^v \\
 1113 \quad &= 316,283,904 \quad (0.316\text{B}).
 \end{aligned}$$

1115 **MLP projector.** The connector maps SigLIP features of width $D_v = 1024$ to the LLM
1116 width $D = 2048$ using

$$1117 \quad f_{\text{proj}}(z) = W_2 \sigma(W_1 z + b_1) + b_2,$$

1118 where $W_1 \in \mathbb{R}^{D \times D_v}$ and $W_2 \in \mathbb{R}^{D \times D}$. Including both biases,
1119

$$\begin{aligned}
 1120 \quad P_{\text{proj}} &= (D_v D + D) + (D^2 + D) \\
 1121 \quad &= (1024 \cdot 2048 + 2048) + (2048^2 + 2048) \\
 1122 \quad &= 6,295,552 \quad (0.006\text{B}).
 \end{aligned}$$

1124 This parameter count is independent of the number of video frames; the same projector is
1125 applied to every visual token.

1126
1127 **Dense Llama-3.2-1B LLM.** For Llama-3.2-1B, let the vocabulary size be $V = 128,256$,
1128 hidden width $D = 2048$, number of decoder layers $L = 16$, FFN width $F = 8192$, query
1129 heads $H_q = 32$, key-value heads $H_{kv} = 8$, and head dimension $d_h = 64$. The key-value pro-
1130 jection width is $D_{kv} = H_{kv} d_h = 512$. The token embedding is tied with the output head, so it
1131 contributes

$$1132 \quad P_{\text{emb}} = VD = 128,256 \cdot 2048 = 262,668,288.$$

1133 Llama attention and FFN projections are bias-free, and RMSNorm contributes one scale
1134 vector per norm. For one decoder layer,

$$\begin{aligned}
 1135 \quad P_{\text{attn}}^\ell &= D(H_q d_h) + 2D(H_{kv} d_h) + D(H_q d_h) \\
 1136 \quad &= 2048^2 + 2 \cdot 2048 \cdot 512 + 2048^2 = 10,485,760, \\
 1137 \quad P_{\text{ffn}}^\ell &= DF + DF + FD = 3DF = 3 \cdot 2048 \cdot 8192 = 50,331,648, \\
 1138 \quad P_{\text{norm}}^\ell &= 2D = 4,096, \\
 1139 \quad P_{\text{layer}}^\ell &= P_{\text{attn}}^\ell + P_{\text{ffn}}^\ell + P_{\text{norm}}^\ell = 60,821,504.
 \end{aligned}$$

1142 The final RMSNorm adds $D = 2048$ parameters. Therefore the dense LLM has
1143

$$\begin{aligned}
 1144 \quad P_{\text{dense}} &= P_{\text{emb}} + L P_{\text{layer}}^\ell + D \\
 1145 \quad &= 262,668,288 + 16 \cdot 60,821,504 + 2048 \\
 1146 \quad &= 1,235,814,400 \quad (1.236\text{B}).
 \end{aligned}$$

1147 The FFN portion of one full dense Llama-3.2-1B decoder copy is
1148

$$1149 \quad P_{\text{FFN}} = L P_{\text{ffn}}^\ell = 16 \cdot 50,331,648 = 805,306,368 \quad (0.805\text{B}).$$

MoTE LLM conversion. Following sparse LVLM reporting [29], we track total and active parameters. Let T be the number of routed task experts, S the number of shared experts, and k the number of active routed task experts for one sample.

The dense FFN already present in P_{dense} is replaced by $T + S$ FFN copies, so the fully merged LLM counts are

$$\begin{aligned} P_{\text{total}}(T, S) &= P_{\text{dense}} + (T + S - 1)P_{\text{FFN}}, \\ P_{\text{active}}(k, S) &= P_{\text{dense}} + (k + S - 1)P_{\text{FFN}}. \end{aligned}$$

For the COIN model, $T = 5$, $S = 1$, and $k = 1$. Hence

$$\begin{aligned} P_{\text{LLM, total}} &= 1,235,814,400 + 5 \cdot 805,306,368 \\ &= 5,262,346,240 \quad (5.262\text{B}), \\ P_{\text{LLM, active}} &= 1,235,814,400 + 805,306,368 \\ &= 2,041,120,768 \quad (2.041\text{B}). \end{aligned}$$

Adding the visual encoder and projector gives the complete merged video-language model sizes:

$$\begin{aligned} P_{\text{VLM, dense}} &= P_{\text{vis}} + P_{\text{proj}} + P_{\text{dense}} = 1,558,393,856 \quad (1.558\text{B}), \\ P_{\text{VLM, total}} &= P_{\text{vis}} + P_{\text{proj}} + P_{\text{LLM, total}} = 5,584,925,696 \quad (5.585\text{B}), \\ P_{\text{VLM, active}} &= P_{\text{vis}} + P_{\text{proj}} + P_{\text{LLM, active}} = 2,363,700,224 \quad (2.364\text{B}). \end{aligned}$$

Thus Table 3 reports LLM-only rounded values for VideoLLM-MoTE-1B+5E: 5.3B total and 2.0B active. Including the SigLIP 2 visual encoder and MLP projector gives 5.6B total and 2.4B active.

A.2 Training Configuration and Route Descriptions

Table 10 lists the main training and evaluation settings for the VideoLLM-MoTE experiments, while Table 11 records the natural-language route descriptions used by the optional prompt-conditioned expert selector. Keeping these two tables together makes the training setup and routing metadata explicit before the prompt examples and evaluation details.

For COIN, the tasks are trained as offline answer-generation problems: visual-frame positions are masked from the streaming silence loss and supervision covers the assistant answer plus EOS. Ego4D narration keeps the streaming silence term, so the narration route learns to produce short narrations at annotated response points and to remain silent elsewhere in the stream. The descriptions in Table 11 are not prompts shown to the decoder; they define the optional prompt-conditioned selection component described in Section 3.2. In the main COIN results, task labels provide the route assignment directly.

A.3 Data Sampling Strategy

The routed MoTE decoder activates a task expert according to the expert id attached to each training sample. The expert buckets are not equally sized: for example, step recognition, next-step forecasting, task recognition, and procedure forecasting can contribute different numbers of training examples. A uniform-over-experts schedule would repeatedly recycle smaller buckets and overtrain their experts, while larger buckets would be under-sampled relative to the evidence available for those tasks. Conversely, unconstrained mixed shuffling can give different data-parallel ranks different routed experts at the same local step.

Hyper-parameter	Value
frame_fps / max_frames	2 / 1200
GPUs	4×A100-80GB
per_device_batch	1 train / 1 eval
gradient_acc_steps	8
Effective train batch	32 samples
num_train_epochs	10
learning_rate	1e-4
optimizer	AdamW
lr_scheduler_type	cosine
warmup_ratio	0.05
lora_targets	q/k/v/o, gate/up/down, lm_head
finetune_modules	MLP projector
lora_r / lora_alpha	128 / 256
lora_dropout	0.05
moe_router_aux_loss	0.0
attn_implementation	sdpa

Table 10: Hyper-parameters used for training VideoLLM-MoTE.

Task	Expert natural-language description
Current Step	Recognize the action currently being performed in the observed video segment; use this route for prompts asking what is happening now or which procedural step is visible.
Next Step	Predict the immediate next action that should follow the observed segment; use this route for short-horizon forecasting prompts asking what comes next.
Task	Identify the overall instructional goal or activity category of the video, independent of the current substep; use this route for prompts asking for the complete task being performed.
Procedure	Forecast a sequence of upcoming procedural actions from the current visual state without being given an explicit task goal; use this route for open-ended procedure-continuation prompts.
Task Procedure	Forecast upcoming procedural actions while conditioning on a provided goal or task name; use this route when the prompt states the target activity and asks for future steps.
Narration	Generate concise timestamp-aligned egocentric narrations for streaming video; use this route for online narration prompts that require deciding both what to say and when to say it.

Table 11: Task-to-description mapping used for prompt-conditioned expert selection.

Figure 3 illustrates the bucketed rank-aligned strategy used in our training runs. At the start of an epoch, samples are grouped by their expert id and each bucket is shuffled. At each local step, the sampler chooses one expert bucket with probability proportional to the number of fresh samples still unseen in that bucket. It then draws a global group of $G = bR$ indices from that selected bucket, where b is the per-rank local batch size and R is the number of data-parallel ranks. Distributed sharding splits this contiguous group across ranks, so every rank receives a local microbatch for the same routed expert at that step. When a selected bucket is exhausted, it is reshuffled and reused only as needed. This preserves the task-size bias of the training set while reducing both overexposure of small buckets and underexposure of large buckets.

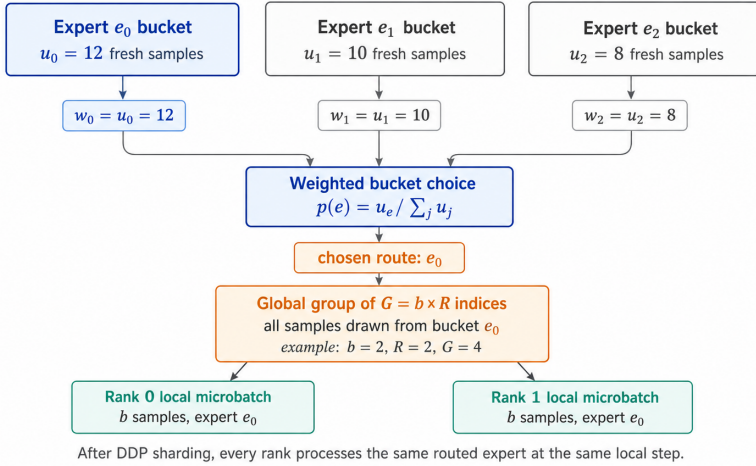


Figure 3: **Bucketed rank-aligned sampling.** The sampler groups a concatenated multi-task dataset into expert buckets. At each local training step, it samples one expert bucket with probability proportional to the number of fresh samples remaining in that bucket, draws one global group of $G = bR$ indices from the selected bucket, and lets distributed sharding split the group into rank-local microbatches. Here b is the local batch size and R is the number of data-parallel ranks.

A.4 Prompt Templates

This section shows the rendered chat strings produced by `apply_chat_template`. The blue text marks visual placeholders and the orange text marks supervised assistant response spans during training. In the live1+ setting, each frame contributes ten repeated `<v>` markers and adjacent frames are comma-separated inside stream brackets; long streams are abbreviated with ellipses for readability. At evaluation time, the assistant target is omitted and the template ends with `Assistant:`.

For COIN, the template serializes the system message, user prompt, video stream, and assistant answer in that order. The shared system message is written once below; the shorthand `[system prompt]` in each prompt expands to this text. The concrete targets below use official annotation labels from a `Make french fries` video. For Ego4D narration, the stream remains online: an initial frame chunk is followed by the narration instruction,

then additional frame chunks and timestamped assistant narrations.

System Prompt	applies to all routes
A multimodal AI assistant is helping users with some activities. Below is their conversation, interleaved with the list of video frames received by the assistant.	
COIN Step Recognition	Step route
< begin_of_text > [system prompt] User: What is the action in the video? Format your answer concisely. No extra text output. Assistant: <i>Cut potato into strips.</i> < eot_id >	
COIN Next-Step Forecasting	Next route
< begin_of_text > [system prompt] User: What is the next action for the video? Format your answer concisely. No extra text output. Assistant: <i>Soak them in water.</i> < eot_id >	
COIN Task Recognition	Task route
< begin_of_text > [system prompt] User: What is the overall activity in the video? Format your answer concisely. No extra text output. Assistant: <i>Make french fries.</i> < eot_id >	
COIN Procedure Forecasting	Proc. route
< begin_of_text > [system prompt] User: What is the next 3 actions for the video? Format your answer concisely, listing each action on a new line with a number prefix. No extra text output. Assistant: <i>1. Soak them in water.</i> <i>2. Dry strips.</i> <i>3. Put in the oil to fry.</i> < eot_id >	
COIN Procedure Forecasting with Task Goal	Proc.+ route
< begin_of_text > [system prompt] User: To <i>Make french fries</i> , what is the next 3 actions for the video? Format your answer concisely, listing each action on a new line with a number prefix. No extra text output. Assistant: <i>1. Soak them in water.</i> <i>2. Dry strips.</i> <i>3. Put in the oil to fry.</i> < eot_id >	
Ego4D Narration Stream	Narration route
< begin_of_text > [system prompt] [<v>...<v>] User: Please concisely narrate the video in real time. Use the tag 'C' to denote the camera wearer, and other letter tags, such as 'X', to denote other individuals in the scene. [<v>...<v>,<v>...<v>] Assistant: <i>C walks around a room.</i> < eot_id > [<v>...<v>,<v>...<v>,<v>...<v>] Assistant: <i>C picks up a wire from the floor.</i> < eot_id >	

A.5 Evaluation Protocols

COIN benchmarks. COIN outputs are decoded as free-form text and then mapped back to the task label space before scoring. For Step, Next, and Task, we lowercase the response,

Method	COIN Benchmark Top-1 Accuracy↑				
	Step	Next	Proc.	Task	Proc.+
VideoLLM-MoTE-1B+5E	65.1	50.3	50.5	94.4	54.2
Swap Step↔Next	21.5	17.4	50.5	94.4	54.2
Swap Next↔Proc.	65.1	23.3	30.7	94.4	54.2
Swap Proc.↔Task	65.1	50.3	4.97	15.36	54.2
Swap Task↔Proc.+	65.1	50.3	50.5	15.98	5.36

Table 12: Offline expert-swap routing diagnostic on COIN. Highlighted cells indicate the two task columns whose expert weights are swapped at evaluation time. Untouched task routes keep the base route weights.

remove a trailing period, and first check for an exact match with the target label. If no exact match is found, the response is assigned to the closest valid class in the task taxonomy using Levenshtein distance. Top-1 accuracy is then computed over the mapped labels. For Proc. and Proc.+, the generated numbered list is split into individual lines, optional number prefixes are removed, and each predicted future step is mapped to the nearest COIN step label before computing step-wise top-1 accuracy over aligned future-action positions.

Ego4D narration stream. Ego4D narration is evaluated as an online stream rather than as an offline classification problem. Each narration turn has an annotated response time and a short target narration. We report three diagnostics following the prior online VideoLLM protocol: PPL measures token-level narration likelihood, TimeDiff measures the absolute temporal offset between the predicted and annotated response time in seconds, and Fluency combines timing and token correctness into a percentage score. Lower PPL and TimeDiff are better, while higher Fluency is better. These metrics are useful for timestamped short narrations, but they should not be read as a complete substitute for human evaluation of open-ended video dialogue.

Receipt KIE and OCR preservation. The auxiliary document experiment is evaluated separately from the video benchmarks. For SROIE Task 3, we report strict field-level F1 for company, date, address, and total, together with micro-F1 over the receipt fields in Table 13. For CORD, we report relaxed field-level F1 over the receipt categories used in the benchmark table and the corresponding micro-F1 in Table 14. The OCR route is checked on OmniDocBench v1.5 using the original GLM-OCR metrics: overall score, text edit distance, formula score, table score, table-structure score, and reading-order edit distance. These document metrics are used to test route adaptation and route preservation; they are not averaged with the video-language results.

A.6 Offline Expert-Swap Routing

Table 12 provides an offline diagnostic for whether COIN task behavior is localized in the routed experts. We keep the evaluation prompts and route ids unchanged, but swap the trained FFN weights of two task experts across all decoder layers before evaluation. If the experts behave as task-specialized modules, the degradation should concentrate on the swapped tasks while the untouched routes remain close to the base model.

The diagnostic shows that accuracy collapses primarily on the swapped task pair. Swapping Step and Next damages only those two columns, while Proc., Task, and Proc.+ remain at the base values. The same locality appears for the other pairs, with the strongest failures occurring when Proc. is swapped with Task or when Task is swapped with Proc.+. This pattern supports the interpretation that routed FFN experts capture task-specific transformations rather than acting as interchangeable capacity. The diagnostic is not an expert-removal or unlearning evaluation; it is a stress test of expert specialization under deliberately incorrect route-to-weight assignments.

A.7 Auxiliary Document Flow Architecture

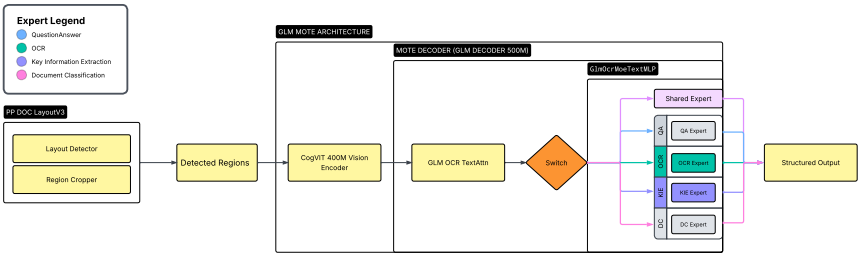


Figure 4: **Auxiliary document-flow architecture.** The auxiliary document experiment adapts the GLM-OCR architecture with MoTE task routes. OCR and receipt-KIE routes are evaluated in this paper; gray routes denote additional document-task capacity that is not included in the reported benchmark numbers.

A.8 Auxiliary Document KIE Protocol and Full Comparisons

This section expands the compact document-domain tables from Section 4.5. The GLM-OCR-MoTE-0.9B+2E conversion uses the same high-level pattern as the video model: an always-active shared FFN path is paired with one selected task route. In this auxiliary setting, the two routed experts correspond to OCR and receipt KIE. During KIE adaptation, the new receipt route learns to emit structured receipt fields directly from image inputs, while the retained OCR route is evaluated separately on the original OCR benchmark.

The comparison groups baselines by input modality because this determines how much document structure is provided to the model. OCR-text-only and OCR-plus-layout baselines receive recognized text, bounding boxes, or both. Image-using baselines receive pixels, sometimes together with OCR or layout. GLM-OCR-MoTE-0.9B+2E is listed as an image-input model because the KIE route predicts fields from the receipt image without an external OCR-text prompt. Tables 13 and 14 report the full per-field values behind the compact main-paper comparisons.

Method	Input Modality	SROIE Task 3 Strict F1 (%) ↑ (KIE Domain)					OmniDocBench v1.5 (Original Task)					
		Comp.	Date	Addr.	Total	Micro	Over.↑	TxtEd.↓	Form.↑	Tab.↑	Tab-S↑	RdEd.↓
Text-Only Baselines (OCR Prompted)												
GPT-4 + OCR [10]	OCR Text Only	-	-	-	-	90.60	-	-	-	-	-	-
Llama2-7B-chat [10]	OCR Text Only	-	-	-	-	68.97	-	-	-	-	-	-
Vicuna-1.5-7B [8]	OCR Text Only	-	-	-	-	68.20	-	-	-	-	-	-
Llama2 + OCR [10]	OCR Text Only	-	-	-	-	56.40	-	-	-	-	-	-
Llama2-7B [10]	OCR Text Only	-	-	-	-	34.96	-	-	-	-	-	-
Spatial / Layout-Aware Baselines (No Image)												
DocLLM-7B [10]	OCR Text + BBoxes	-	-	-	-	91.90	-	-	-	-	-	-
Multimodal Baselines (Image + Text)												
LayoutLLM-7B [10]	Image+Text+Layout	-	-	-	-	72.12	-	-	-	-	-	-
DocKyllin [10]	Image + Text	-	-	-	-	69.60	-	-	-	-	-	-
Qwen-VL-7B [9]	Image + Text	-	-	-	-	58.59	-	-	-	-	-	-
LLaVA-1.5-7B [10]	Image + Text	-	-	-	-	3.83	-	-	-	-	-	-
LLaVAR-7B [10]	Image + Text	-	-	-	-	2.38	-	-	-	-	-	-
Base OCR Architecture												
GLM-OCR-0.9B [10]	Image	49.76	78.22	17.81	74.40	55.04	94.62	0.040	93.90	93.96	96.39	0.044
Mixture of Task Experts Adaptation (Ours)												
GLM-OCR-MoTE-0.9B+2E (ours)	Image	90.49	97.41	70.03	93.66	87.90	94.62	0.040	93.90	93.96	96.39	0.044

Table 13: Full SROIE and OmniDocBench comparison corresponding to Table 6. The first metric block evaluates receipt KIE, while the second checks preservation of the original OCR route.

Method	Input Modality	CORD Relaxed F1 Score (%) ↑								
		Cash	Chng	Cred	Menu	Serv	SubTot	Tax	Total	Micro
<i>Text-Only Baselines (OCR Prompted)</i>										
RoBERTa [10]	OCR Text Only	–	–	–	–	–	–	–	–	91.98
BERT_large/T5 [10, 10]	OCR Text Only	–	–	–	–	–	–	–	–	90.25
<i>Spatial / Layout-Aware Baselines (No Image)</i>										
LayoutLM [10]	OCR Text + BBoxes	–	–	–	–	–	–	–	–	94.93
LAMBERT [10]	OCR Text + BBoxes	–	–	–	–	–	–	–	–	94.41
<i>Multimodal Baselines (Image + Text + Layout)</i>										
UDOP [10]	Image+Text+Layout	–	–	–	–	–	–	–	–	97.60
LayoutLMv2 [10]	Image+Text+Layout	–	–	–	–	–	–	–	–	96.01
DocTr [10]	Image+Text+Layout	–	–	–	–	–	–	–	–	94.40
Donut [10]	Image + Text	–	–	–	–	–	–	–	–	91.60
<i>Base Architecture (Before MoTE)</i>										
GLM-OCR-0.9B [10]	Image	11.43	20.59	16.67	0.00	47.06	42.55	15.69	28.12	20.07
<i>Mixture of Task Experts Adaptation (Ours)</i>										
GLM-OCR-MoTE-0.9B+2E (ours)	Image	95.45	99.10	93.33	90.00	100.00	100.00	100.00	95.24	95.79

Table 14: Full CORD KIE comparison corresponding to Table 7. GLM-OCR-MoTE-0.9B+2E uses the receipt KIE route and predicts structured fields directly from the receipt image.

A.9 Limitations

MoTE is most effective when the task set has meaningful and stable boundaries. If tasks overlap heavily, or if a prompt underspecifies the intended task, sample-level routing can select an unsuitable expert; softer routing, hierarchical experts, or expert composition may be needed in such cases. Prompt-conditioned expert selection is currently a design mechanism rather than a fully evaluated result: the main COIN experiments use explicit task-index routing, so future work should report selection accuracy and downstream prompt-only performance. Finally, incremental expert addition protects existing routes by freezing old modules, but this reduces plasticity for new tasks; stronger expansion may require controlled updates to shared representations or rehearsal.